

## LAB 2

### Task # 01 (Creating a Simple Class)

- Create a class named Car.
- Define instance variables for color, brand, and model.

#### **CODE:**

```
class Test {  
  
    public static void main(String[] args) {}  
  
}  
  
public class Car {  
  
    String color, brand , model;  
  
}
```

#### **OUTPUT:**

No output from this code

### Task # 02 (Adding a Constructor)

- Add a constructor to the Car class that initializes the instance variables.
- Create a constructor with no parameters (default constructor).

#### **CODE:**

```
class Test {  
  
    public static void main(String[] args) {}  
  
}  
  
public class Car {  
  
    String color, brand , model;  
  
    Car(){  
  
        this.color = "Red";  
  
        this.brand = "Toyota";  
  
        this.model = "2022";  
  
    }  
  
}
```

#### **OUTPUT:**

No output from this code

# LAB 2

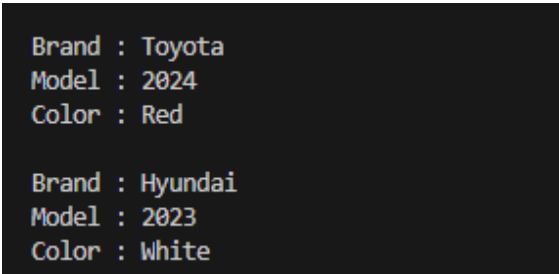
## Task # 03 (Creating Objects)

- Create two Car objects with different values for the instance variables.
- Print the values of the instance variables for each object..

### CODE:

```
class Test {  
  
    public static void main(String[] args) {  
  
        Car car1 = new Car("2024","Red","Toyota");  
  
        car1.show();  
  
        Car car2 = new Car("2023","White","Hyundai");  
  
        car2.show();  
  
    }  
}  
  
public class Car {  
  
    String color, brand , model;  
  
    Car(String model , String color , String brand){  
  
        this.color = color;  
  
        this.brand = brand;  
  
        this.model = model;  
  
    }  
  
    public void show(){  
  
        System.out.println("Brand : "+ this.brand);  
  
        System.out.println("Model : "+ this.model);  
  
        System.out.println("Color : "+ this.color);  
  
    }  
}
```

### OUTPUT:



```
Brand : Toyota  
Model : 2024  
Color : Red  
  
Brand : Hyundai  
Model : 2023  
Color : White
```

# LAB 2

## Task # 04 (Using Getters and Setters)

- Add getter and setter methods for the instance variables.
- Use the setter methods to modify the values of the instance variables for one of the objects.
- Print the updated values of the object.

### **CODE:**

```
class Test {

    public static void main(String[] args) {

        Car car1 = new Car("2024","Red","Toyota");

        car1.setter("2023","Cyan","Lamborghini");

        car1.getter();

        Car car2 = new Car("2023","White","Hyundai");

        car2.getter();

    }

}

public class Car {

    String color, brand , model;

    Car(String model , String color , String brand){

        this.color = color;

        this.brand = brand;

        this.model = model;

    }

    void setter (String model , String color , String brand)
    {

        this.brand = brand;

        this.model = model;

        this.color = color;
```

```
        System.out.println("Values updated successfully");

    }

    void getter () {

        System.out.println("\nBrand : "+ this.brand);

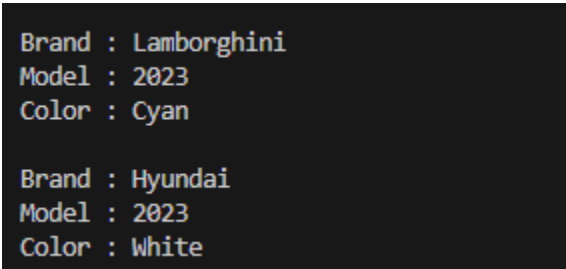
        System.out.println("Model : "+ this.model);

        System.out.println("Color : "+ this.color);

    }

}
```

### **OUTPUT:**



```
Brand : Lamborghini
Model : 2023
Color : Cyan

Brand : Hyundai
Model : 2023
Color : White
```

## LAB 2

### Task # 05 (Creating a Method)

- Add a method to the Car class named startEngine() that prints a message like "Engine started".
- Call the startEngine() method for both objects.

#### **CODE:**

```
class Test {  
  
    public static void main(String[] args) {  
  
        Car car1 = new Car("2024","Red","Toyota");  
  
        car1.startEngine();  
  
        Car car2 = new Car("2023","White","Hyundai");  
  
        car2.startEngine();  
  
    }  
}  
  
public class Car {  
  
    String color, brand , model;  
  
    Car(String model , String color , String brand){  
  
        this.color = color;  
  
        this.brand = brand;  
  
        this.model = model;  
  
    }  
  
    void startEngine(){  
  
        System.out.println("Engine started.");  
  
    }  
}
```

#### **OUTPUT:**

```
Engine started.
```

```
Engine started.
```

## LAB 2